

Hallucination Detection and Evaluation in Multimodal Large Language Models

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Problem

Multimodal LLMs have demonstrated strong capabilities in visual question answering and image-text reasoning, yet they remain susceptible to **hallucinations**: outputs inconsistent with the visual input, contradicting world knowledge, or containing logical flaws in the reasoning process.

Task Requirements:

- ➊ **Definition & classification:** define ≥ 2 hallucination types with explanations, plus ≥ 2 examples per type.
- ➋ **Detection methods & metrics:** build ≥ 2 methods (e.g., rule-based + MLLM-Judge) and design ≥ 2 evaluation metrics.
- ➌ **Experimental evaluation (main task):** evaluate ≥ 2 models (one closed-source, one open-source) on POPE + one recommended reasoning benchmark (MathVista / ChartQA).
- ➍ **Human alignment (Human-as-Judge):** 20–50 human-annotated gold labels; compare auto verdicts vs human judgments, analyze inconsistent cases.
- ➎ **Analysis:** CoT impact on hallucinations, cross-model comparison, and in-depth analysis of ≥ 2 typical failure cases.
- ➏ **Extension:** extend to other image-text tasks or domains (OCR, medical, remote sensing), or perform a cross-task generalization analysis.

Evaluated Models

Model	Role	Architecture	Access
GPT-5.4-mini	Evaluated	Dense	Closed
Gemini 2.5 Flash	Evaluated	Dense	Closed
Qwen3.5-35B-A3B	Evaluated	MoE 35 B	Open
Qwen3-VL-235B-A22B	Evaluated	MoE 235 B	Open
GPT-5.5	MLLM Judge	Dense	Closed
Claude Opus 4.7	Judge consist.	Dense	Closed

Evaluation Datasets

Dataset	Task Type	Samples
POPE	Object existence probing	3,000 (3 × 1,000)
MathVista	Math visual reasoning	1,000 (testmini)
VQA-RAD	Medical radiology VQA	451 (test)

- **POPE** converts object hallucination evaluation into a binary existence probing task built on top of the MS COCO *val2014* split. Each split contains 3,000 probing questions with a balanced 1,500 Yes / 1,500 No ground-truth distribution.
- **MathVista** is a comprehensive benchmark for evaluating the mathematical reasoning of foundation models in visual contexts. The task requires simultaneous visual perception, mathematical knowledge, and multi-step reasoning, well suited to trigger all three hallucination types.
- **VQA-RAD** is the first manually constructed medical VQA dataset in which clinicians generated questions and reference answers directly from radiology images. It serves as a high-stakes domain testbed because both visual perception (anatomy, pathology) and clinical knowledge are required.

Experimental conditions.

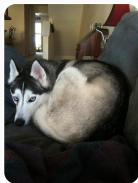
All four models are accessed through APIs with a unified random seed of 42. Temperature is kept at each platform's default, and the maximum token length is uncapped. POPE samples 1,000 items per split; MathVista uses the full `TESTMINI` set (1,000 items); VQA-RAD uses the test split (451 items). GPT-5.5 serves as the MLLM Judge throughout, and a human alignment study is conducted as part of the error analysis.

Definition and Three-Category Taxonomy

Definition

An MLLM hallucination is a model response that is inconsistent with the input visual content, contradicts verifiable real-world knowledge, or contains logical flaws in the reasoning process.

Taxonomy



An image of a husky lying on a couch. Object-level faithfulness hallucinations include category substitution (reporting a cat instead of a dog), object omission (claiming no animal is present), and existence misjudgment (affirming the presence of a non-existent object).

Faithfulness (Visual Inconsistency)

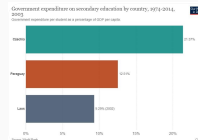
A model response that is inconsistent with the visual content of the input image.



An image showing zebras and giraffes. Commonsense factuality hallucinations arise when the model's roughly correct visual perception leads to outputs that contradict basic biological knowledge, such as misidentifying a zebra as a painted horse.

Factuality (Factual Inconsistency)

A model response that contradicts established, verifiable real-world knowledge.



A bar chart showing government expenditure percentages for Czechia, Paraguay, and Laos. The model correctly reads the bar values but errs in the comparison step, producing a conclusion that contradicts its own extracted evidence.

Logical (Reasoning Error)

A model response with flaws in the reasoning process, where the conclusion does not logically follow from the visual evidence or its own premises.

Detection Methods

Rule-based Method (POPE)

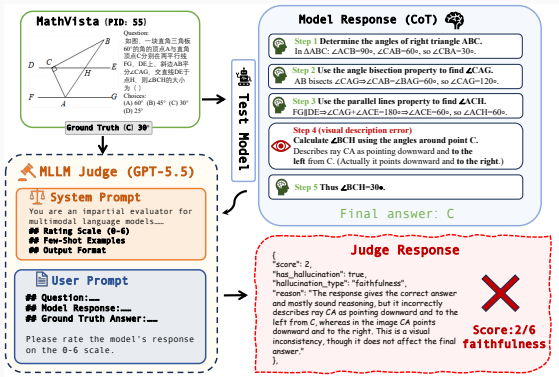
POPE operationalizes object hallucination as a binary probing task: for each image, the benchmark issues yes/no questions “Is there an X in the image?”, and the model commits an object hallucination whenever it answers “yes” while the ground-truth label is “no” (a false positive in the binary-classification view).

Answer normalization (follows the RUCAIBox/POPE official script):

- 1 Take the first sentence of the response (split by “.”).
- 2 Remove all commas; split by whitespace into a set of words.
- 3 If “No”, “not”, or “no” appears in the set, return “no”.
- 4 Otherwise return “yes”.

MLLM Judge (MathVista + VQA-RAD)

Follows the MMHal-Bench evaluation protocol: a 0–6 Likert scale jointly assesses informativeness and hallucination level of each response. A threshold of 3 separates the two regimes: scores below 3 indicate hallucination, scores at or above 3 indicate no hallucination. Whenever the score is below 3, the judge additionally labels the hallucination type (faithfulness, factuality, or logical).



MLLM Judge evaluation framework: the judge receives a question, a model response, and a scoring rubric, and produces a score (0–6) with an optional hallucination type label.

Evaluation Metrics

POPE (rule-based, classification metrics)

Metric	Formula	Description
Accuracy	$(TP + TN)/N$	Overall correctness
Precision	$TP/(TP + FP)$	Acc. when predicting Yes
Recall	$TP/(TP + FN)$	Detection rate for Yes
F1	$2PR/(P + R)$	Harmonic mean of P, R
Yes Ratio	N_{yes}/N	Response-bias indicator
Hallucination Rate	FP/N	Core metric

MathVista (MLLM Judge, 0–6 score)

Metric	Definition	Description
Hallucination Rate	$N_{\text{score} < 3}/N$	Lower is better
Average Score	$\sum \text{score}_i / N$	Response quality
Type-specific	faith./fact./logical	Source breakdown

VQA-RAD (MLLM Judge + medical adaptation)

Metric	Definition	Description
Hallucination Rate	$N_{\text{score} < 3}/N$	Core metric
Average Score	$\sum \text{score}_i / N$	Response quality
Type-specific	faith./fact./logical	Source breakdown
Stratified HR	by closed / open	Answer-type strata

Hallucination Rate (HR) is the core metric across all three datasets, with two operationalizations: a false-positive rate on POPE (rule-based classification view) and the proportion of responses with Judge score < 3 on MathVista and VQA-RAD.

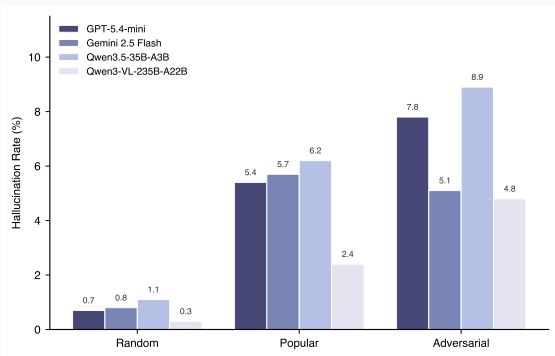
Accuracy measures the proportion of correct predictions ($(TP + TN)/N$ on POPE). It jointly reflects Yes-correct and No-correct, but can be misleading when the model exhibits strong response bias (e.g., answering “yes” on nearly all questions).

Precision is the proportion of Yes predictions that are correct ($TP/(TP + FP)$). High precision means the model rarely imagines non-existent objects. Low precision directly corresponds to a high object hallucination rate.

Recall is the proportion of actual Yes items that are correctly predicted ($TP/(TP + FN)$). High recall ensures the model reliably detects objects that are genuinely present.

Yes Ratio detects response bias on POPE: values near 0.5 signal balanced predictions, a high ratio suggests over-claiming, a low ratio suggests over-rejection.

POPE: Results and Cross-Model Analysis



Hallucination Rate increases from *Random* → *Popular* → *Adversarial*, exposing progressively harder object-existence probes.

Split	Metric	GPT	Gemini	Qwen3.5	Qwen3-VL
Random	HR (%)	0.70	0.80	1.10	0.30
	F1	0.892	0.910	0.922	0.913
	YesR	0.414	0.431	0.447	0.425
Popular	HR (%)	5.40	5.70	6.20	2.40
	F1	0.852	0.858	0.872	0.891
	YesR	0.464	0.475	0.496	0.444
Adversarial	HR (%)	7.80	5.10	8.90	4.80
	F1	0.825	0.870	0.856	0.871
	YesR	0.483	0.474	0.529	0.470

- **Universal difficulty gradient.** All four models show HR rising Random → Popular → Adversarial, confirming POPE's intended progression.
- **Qwen3-VL** has the lowest HR on all splits: strongest hallucination control.
- **Gemini** is most robust from Popular to Adversarial (HR 5.70% → 5.10%).
- **Qwen3.5** maximizes Recall but over-claims (Adversarial YesR = 0.529).
- **GPT-5.4-mini** is conservative; the “default No” shortcut collapses under adversarial pressure.

Three Splits Detect Three Error Mechanisms

1. Random: object-level perception

Negative objects are visually unrelated to the scene. HR remains near saturation (0.30%–1.10%), so remaining errors mainly reflect the visual encoder's parametric prior P_V .

2. Popular: linguistic-prior override

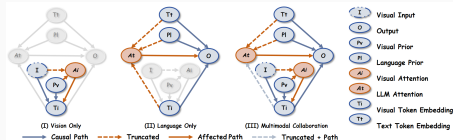
Absent objects are high-frequency co-occurrences (e.g., outdoor eating → backpack). HR jumps to 2.40%–6.20% because the LLM prior P_L suppresses weak or ambiguous visual evidence.

3. Adversarial: semantic-boundary failure

Distractors are near-synonyms of visible objects (e.g., teddy bear vs. bear). HR peaks at 4.80%–8.90% when P_V blurs fine-grained categories and P_L merges them semantically.

The architectural asymmetry of MLLMs.

Mainstream MLLMs connect a CLIP/SigLIP-style vision encoder to a much larger LLM through a projector. Visual tokens are mapped into the word-embedding space, then mixed with text tokens inside the LLM's self-attention. Because the language backbone has far more parameters and pretraining data than the visual pathway, language priors have a structural advantage whenever visual evidence is weak or fine-grained.



Two independent confounding factors

Visual modality prior P_V . The vision encoder carries biases from CLIP/SigLIP pretraining. It determines which objects, textures, and fine-grained category boundaries are easy or difficult to encode. In Random, this prior mostly measures baseline visual perception; in Adversarial, it blurs near-synonyms such as “teddy bear” and “bear.”

Linguistic modality prior P_L . The LLM backbone carries co-occurrence regularities from text-only pretraining. When an image context suggests a likely object but the object is absent, this prior can activate a plausible answer and overpower weak visual evidence. Popular isolates this effect most clearly, while Adversarial combines it with visual semantic confusion.

POPE Case Analysis: Two Systematic False Positives



Case 1: Adversarial split

Present: teddy bear, person, book
Query: "Is there a **bear** in the image?"

Ground truth: **No**

Model	GPT	Gemini	Qwen3.5	Qwen3-VL
Answer	Yes	Yes	Yes	Yes

Case 1: semantic-boundary confusion

The image contains a *teddy bear*, but POPE asks about *bear* as a real animal category. All models merge the toy category with the animal category, showing that the adversarial split tests fine-grained conceptual boundaries, not only low-level visual resolution.



Case 2: Popular split

Present: person, sandwich, hot dog
Query: "Is there a **backpack** in the image?"

Ground truth: **No**

Model	GPT	Gemini	Qwen3.5	Qwen3-VL
Answer	Yes	Yes	Yes	Yes

Case 2: linguistic-prior override

In outdoor eating scenes, *backpack* is a plausible high-frequency co-occurring object. When the visual signal is weak or cluttered, the language prior fills in a statistically likely object that is absent from the actual image.

MathVista Results and Hallucination Distribution

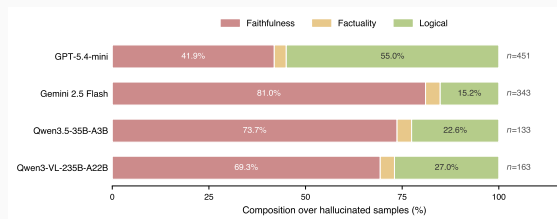
Result summary: Direct vs. CoT prompting

Model	Hallucination Rate (%)			Average Score		
	Direct	CoT	Δ	Direct	CoT	Δ
GPT-5.4-mini	45.1	34.4	-10.7	3.31	4.35	+1.04
Gemini 2.5 Flash	34.3	27.6	-6.7	4.46	4.81	+0.35
Qwen3.5-35B-A3B	13.3	15.5	+2.2	5.00	5.34	+0.34
Qwen3-VL-235B	16.3	24.0	+7.7	4.85	4.94	+0.09

Model	Δ by Hallucination Type			Total
	Faith.	Fact.	Logical	Δ
GPT-5.4-mini	+8	-10	-105	+107
Gemini 2.5 Flash	-66	-8	+7	+67
Qwen3.5-35B-A3B	+26	-3	-1	-22
Qwen3-VL-235B	+41	0	+36	-77

Key observations: CoT polarization

- **Weak baselines improve** — GPT-5.4-mini (-10.7 pp, mainly logical 248 $\rightarrow$ 143); Gemini (-6.7 pp, mainly faithfulness 278 $\rightarrow$ 212).
- **Strong baselines degrade** — Qwen3.5 (+2.2 pp, faithfulness 98 $\rightarrow$ 124); Qwen3-VL (+7.7 pp, faithfulness 113 $\rightarrow$ 154; 116 Direct-correct samples turn hallucinated).
- Average score rises for all four models, yet HR rises for both Qwens: CoT trades terseness for exposable reasoning.

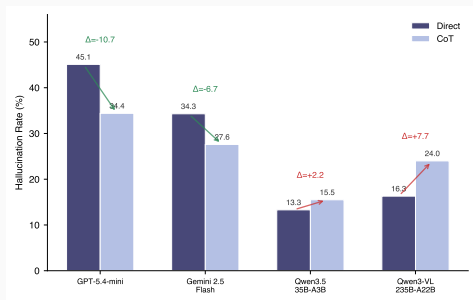


Hallucination-type composition across models and prompting modes.

Distribution differences

- **Faithfulness dominates 7/8 configurations** (> 57% each), peaking at Gemini Direct (81.0%). MathVista's bottleneck: "see first" (numbers, labels, arrows, spatial relations) before reasoning.
- **Exception: GPT-5.4-mini Direct** (logical 55.0%). Median response is only one word, so wrong final options are judged as logical failures rather than visible reasoning mistakes. CoT recovers the normal distribution (logical $\rightarrow$ 41.6%).
- **Factuality stays < 4%** in all configurations. The main bottleneck is visual grounding plus reasoning, not missing world knowledge.

CoT Impact: Results and Four Mechanisms



Observed impact

CoT helps weaker Direct baselines: GPT-5.4-mini HR drops from 45.1% to 34.4% (−10.7 pp), and Gemini drops from 34.3% to 27.6% (−6.7 pp). It hurts stronger Direct baselines: Qwen3.5 rises to 15.5% (+2.2 pp), and Qwen3-VL rises to 24.0% (+7.7 pp).

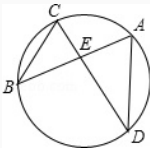
Key tension: Average score increases for all four models, but hallucination rate increases for both Qwen models. CoT can make answers more informative while also exposing judge-detectable hallucinations inside the reasoning chain.

Four mechanisms

The results suggest that CoT both reduces and introduces hallucinations through four simultaneous mechanisms. The first two explain improvements on weaker models, whereas the last two explain degradation on stronger models.

- 1 **Structured decomposition.** CoT decomposes a complex multi-step problem into controllable intermediate steps. When the model is forced to derive rather than guess, many problems that cannot be solved in a single step become answerable.
- 2 **Visual grounding.** CoT prompts require the model to explicitly cite visual evidence during reasoning, reducing its tendency to take linguistic-prior shortcuts. The reasoning constraint pushes the model to inspect the chart before describing it.
- 3 **Exposure effect.** Visual perception errors that remain hidden in Direct mode are explicitly described in the CoT reasoning chain and captured by the MLLM Judge. Even when the final answer is correct, visual misreadings in intermediate steps lower the score.
- 4 **Snowball effect.** CoT forces the model to make explicit visual or mathematical assertions early in the reasoning process. Once the model produces a hallucination early in reasoning, it elaborates coherently around the erroneous premise, causing errors to propagate and amplify along the reasoning chain.

MathVista Case Analysis: Snowball vs Exposure



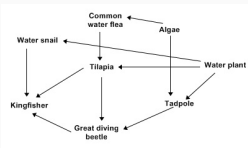
Case 1: CoT snowball

Geometry reasoning: two chords AB , CD intersect at E ;
 $\angle D = 35^\circ$. Find $\angle C$.
Ground truth: **B** (70°)

Mode	Response	Score	Type
Direct	B (70°)	5	none
CoT	flawed reasoning $\rightarrow$ C (80°)	1	logical

Case 1 : Snowball effect

In Direct mode, the model directly outputs the correct option. In CoT mode, it misapplies the triangle angle-sum formula in the first step. After finding that the computed answer (40°) is not among the options, it fabricates a supplementary-angle explanation and arrives at 80° . Once an incorrect premise is locked in, subsequent steps appear plausible but rest on the initial false assumption.



Case 2: CoT exposure

Biology food web reasoning: which organism is most affected if algae is eliminated?
Ground truth: **B** Common water flea

Mode	Response	Score	Type
Direct	B (correct answer)	5	none
CoT	B + false arrow claim	2	faith.

Case 2 : Exposure effect

In CoT mode, the model gives a detailed food-web analysis but incorrectly claims that “Algae has arrows pointing to Tilapia.” The diagram contains no such arrow. Direct mode may still reach the correct answer through a shortcut or partial visual perception; CoT forces the model to describe arrow relationships, exposing the visual misreading to the MLLM Judge and dropping the score from 5 to 2.

Overall Results				
Metric	GPT-5.4-mini	Gemini 2.5 Flash	Qwen3.5-35B	Qwen3-VL-235B
Hallucination Rate	33.26%	66.30%	39.25%	58.31%
Average Score	4.52	2.68	4.20	3.57
$n_{\text{hallucinated}}$	150	299	177	263
Faithfulness	145 (96.7%)	296 (99.0%)	171 (96.6%)	257 (97.7%)
Factuality	1 (0.7%)	2 (0.7%)	4 (2.3%)	5 (1.9%)
Logical	4 (2.7%)	1 (0.3%)	2 (1.1%)	1 (0.4%)
None	301	152	274	188

Result analysis

- **Faithfulness dominates absolutely:** 96.6%–99.0% of all hallucinations are faithfulness errors, so medical VQA failures mainly come from weak radiological visual grounding rather than missing medical knowledge.
- **Model ranking reverses across tasks:** GPT-5.4-mini performs best here (HR = 33.26%, Avg = 4.52), while Qwen3-VL, strongest on POPE, regresses to HR = 58.31%. Natural-image competence does not transfer reliably to X-ray/MRI perception.
- **Open questions are harder than closed ones:** every model shows higher HR on open-ended answers, indicating that lesion localization and free-text description expose the knowledge–perception gap more strongly than conservative yes/no screening responses.



Case 1: Conservative refusal

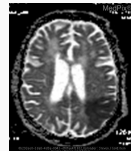
Chest X-ray: evidence of an aortic aneurysm?

Expected answer: **Yes**

Model	Summary	Score	Type
GPT-5.4	no convincing evidence	2	faith.
Gemini	cannot confirm	2	faith.
Qwen3.5	suggests aneurysm	6	none
Qwen3-VL	no clear evidence	2	faith.

Case 1 : Conservative refusal

Three models describe abnormal signs such as mild widening or prominence of the mediastinal aorta, but ultimately refuse to make a positive diagnosis. They perceive the abnormality but retreat at the decision stage. In clinical screening, this conservative answer is equivalent to a missed diagnosis.



Case 2: Knowledge $\neq$ perception

Axial MRI: which side is the lesion?

Expected answer: **left** (radiologist's view)

Model	Summary	Score	Type
GPT-5.4	reverses convention	2	faith.
Gemini	reads MRI as abdominal CT	0	faith.
Qwen3.5	mislocalizes lesion	2	faith.
Qwen3-VL	wrong right-frontal claim	2	faith.

Case 2 : Faulty vision

The lesion sits on the viewer's right, corresponding to the patient's left hemisphere. GPT-5.4-mini sees the image-side location but applies the convention in the wrong direction. The Qwen models recite the rule correctly but mislocalize the lesion. Fluent clinical terminology fills the gap left by uncertain visual evidence.

Human Alignment Verification

Setup

- 40 MathVista CoT samples, stratified by model (10 per model; seed 42).
- Single-blind annotation: model name and Judge verdict hidden.
- Human annotator labels hallucination presence and type.
- Metrics: Accuracy, Precision, Recall, F1, Cohen's κ , and type agreement.

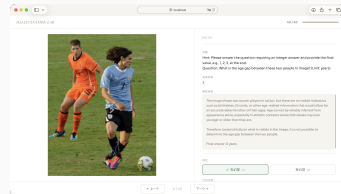
Key results (per-model breakdown)

Subset	N	Accuracy	F1	κ
Overall	40	0.850	0.824	0.700
GPT-5.4-mini	10	1.000	1.000	1.000
Qwen3.5-35B	10	1.000	1.000	1.000
Qwen3-VL-235B	10	0.800	0.750	0.600
Gemini 2.5 Flash	10	0.600	0.333	0.200

Overall Recall = 1.000 (no misses); Type agreement = 0.929 (13/14).

Confusion matrix (Human \ Judge)

Human \ Judge	No halluc.	Halluc.	Total	
No halluc.	20	6	26	← over-judge
Halluc.	0	14	14	← no miss
Total	20	20	40	



Annotation interface used in the single-blind human alignment study.

Analysis conclusions

- **Substantial agreement:** the Judge reaches 85.0% accuracy, $F1 = 0.824$, and $\kappa = 0.700$.
- **No missed hallucinations:** recall is 1.000, so all 14 human-flagged hallucinations are also caught by the Judge.
- **Disagreements are one-sided:** all 6 errors fall into the human-no / Judge-yes quadrant, making over-judgment the dominant failure mode.
- **Over-judgment pattern:** mismatches concentrate on unobservable inference such as age gaps or birth decades; the model refuses to guess, while the Judge still penalizes the mismatch with the reference.
- **Type labels are reliable:** type agreement reaches 0.929 (13/14), showing that the three-way taxonomy remains stable once hallucination is detected.

Error Experiments I: Judge and Threshold Robustness

1. Judge model consistency

Setup: 100 MathVista samples per response model; GPT-5.5 and Claude Opus 4.7 use identical prompts.

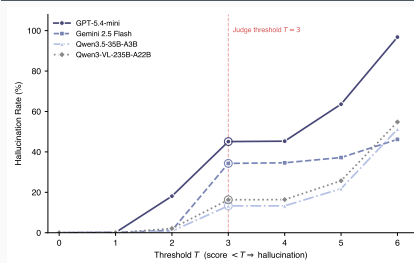
Model	GPT HR	Claude HR	Δ HR	κ	ρ
GPT-5.4-mini	45.0%	44.0%	-1.0	0.980	0.805
Gemini 2.5	37.0%	27.0%	-10.0	0.591	0.658
Qwen3.5	14.0%	18.0%	+4.0	0.777	0.722
Qwen3-VL	15.0%	22.0%	+7.0	0.638	0.669
Overall	-	-	-	0.776	0.740

- **Substantial overall agreement** ($\kappa = 0.776$, $\rho = 0.740$). Model ranking is fully preserved under both judges: Qwens > GPT-5.4-mini > Gemini.
- **GPT-5.4-mini is the most stable** ($\kappa = 0.98$). Its often single-token Direct responses leave almost no room for judge interpretation.
- **Gemini diverges the most** ($\kappa = 0.591$; Claude rates HR 10 pp lower than GPT-5.5). Verbose responses sit closer to the “informative but wrong” calibration boundary between the two judges.
- **Limitation:** both judges are frontier closed-source models; an open-source or domain-specific judge would strengthen the claim.

2. Threshold sensitivity

$HR(T)$ is recomputed as $N_{\text{score} < T} / N$ for MathVista Direct, varying the judge-score threshold.

Model	$T=2$	$T=3$	$T=4$	$\Delta(3 \rightarrow 2)$	$\Delta(4 \rightarrow 3)$
GPT-5.4-mini	18.1%	45.1%	45.3%	-27.0	+0.2
Gemini 2.5	1.2%	34.3%	34.6%	-33.1	+0.3
Qwen3.5	0.8%	13.3%	13.3%	-12.5	0.0
Qwen3-VL	2.1%	16.3%	16.4%	-14.2	+0.1



- $HR(T=3)$ and $HR(T=4)$ differ by < 0.3 pp: scores are strongly bimodal.
- The largest jump is $T=2 \rightarrow 3$ (12.5–33.1 pp); ranking remains unchanged.

Error Experiments II: Length and Cross-Model Consistency

3. Length bias analysis

Spearman correlations test whether longer MathVista responses receive systematically higher judge scores.

Model	Direct ρ	CoT ρ	Direct Q1→Q4	CoT Q1→Q4
GPT-5.4-mini	+0.086	+0.015	3.19→3.46	4.17→3.95
Gemini 2.5	-0.142	-0.218	4.61→3.80	5.30→4.33
Qwen3.5	+0.530	+0.006	4.56→5.33	5.32→5.26
Qwen3-VL	+0.394	-0.241	4.38→4.92	5.32→4.32

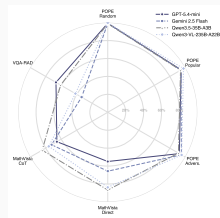
- **No uniform length bias.** Correlation sign depends on each model's quality pattern; there is no systematic "longer → higher score".
- **Qwen3.5 Direct $\rho = +0.530$** (panel maximum): Q1→Q4 of 4.56→5.33 shows "longer because more correct", not "higher because longer".
- **Gemini negative correlation** (-0.142): verbose responses sit closer to the "informative but wrong" boundary, so length itself predicts a lower score.
- **CoT reverses positive correlations** (Qwen3-VL: +0.394 → -0.241): longer chains contain more visual assertions and therefore more opportunities to be penalized.

4. Cross-model hallucination consistency

For each item, count how many of the four models hallucinate. Hard Ratio = samples where $\geq 3/4$ models hallucinate.

Dataset	0/4	1/4	2/4	3/4	4/4	Hard
MathVista (Direct)	387	321	175	49	68	11.7%
MathVista (CoT)	487	232	132	77	72	14.9%
VQA-RAD	59	118	122	81	71	33.7%

- **VQA-RAD difficulty is intrinsic.**
Only 13.1% fully correct (59/451);
33.7% with $\geq 3/4$ hallucinating.
High HR is a domain-level perception gap, not an idiosyncratic model weakness.
- **CoT polarization signature.**
Direct: 38.7% fully correct / 6.8% fully wrong; CoT pushes both tails to 48.7% / 7.2%, middle cells shrink.
- **No single dominant model.** Each model has a different weak axis across POPE / MathVista / VQA-RAD; cross-task trade-offs persist.



Radar: 6 dims.

① Narrow benchmark coverage

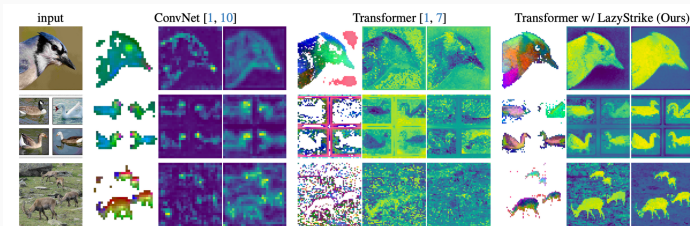
- Three datasets (POPE, MathVista, VQA-RAD) probe three distinct hallucination regimes but do not span the full landscape.
- Excluded: HallusionBench (visual illusions and language-context manipulations), AMBER (multi-dimensional, LLM-free), MM-Vet and MMBench (broader capability suites).
- Whether the CoT polarization effect and the knowledge–perception gap on VQA-RAD generalize to these probes remains untested.

② Restricted model panel

- Four models from three families (OpenAI, Google, Alibaba).
- Smaller open-weight models (LLaVA, MiniGPT, InstructBLIP families) and parameter-scale ablations within a single family are excluded.
- Cannot disentangle effects driven by model scale from those driven by training methodology or visual-encoder choice.

③ Sample-limited human alignment

- 40-sample annotation pool yields $\kappa = 0.70$ with 95% CI [0.48, 0.90] — wide enough for aggregate “substantial” agreement but not for per-model claims.
- The per-model breakdown (Gemini $\kappa = 0.20$, Qwen3-VL $\kappa = 0.60$) is suggestive but each model is evaluated on only 10 samples.
- A scaled-up effort ($N \geq 80$ per model) would be required for statistically grounded cross-model claims.



Vanilla ViT scatters attention across the background; LAST-ViT concentrates on the foreground subject. Reproduced from Shi et al. (2026), Fig. 1.

LAST-ViT (Shi et al., CVPR 2026)

- In standard ViTs, the CLS token (image-level summary used for final prediction) tends to draw most of its content from **background** regions rather than the actual subject.
- Reason is mundane: background patches occupy more area, and the training objective only checks the final label without constraining *which patches* the model attends to.
- Fix: lightweight plug-in that re-weights how patch information aggregates into CLS, **biasing it toward the foreground without retraining the backbone** (consistent gains across twelve vision benchmarks).

Parallel insight → MLLM hallucination

- Same shortcut pattern, different axis: instead of *background patches drowning out foreground patches in one image*, the *language prior drowns out the image signal across a conversation turn*.
- CausalMM's dual-prior framework gives this a causal grounding: P_V and P_L act through independent confounding pathways, and intervention on either can partially offset the other.
- **Conjecture:** A decoding-time re-weighting strategy that pulls attention back toward visual tokens when P_V and P_L conflict may reduce language-prior hallucinations *without degrading language capability*.

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Thank You

Code & data:

<https://github.com/lexiangrui/Hallucination-of-MLLM>

Questions & discussion welcome